

IoT

IoT Protocol Stack

1. [Application Layer] “A comparative analysis and implementation of CoAP and MQTT protocol for IoT communication,” *Life Cycle Reliability and Safety Engineering*, Springer (2025).
2. [Application Layer] Comparative study of MQTT, CoAP, and OPC UA on the FIT-IoT testbed, *Applied Sciences*, MDPI, vol. 11, no. 11 (2023).
3. [Network Layer] “A comprehensive survey on RPL routing-based attacks, defences and future directions in Internet of Things,” *Computer Communications*, ScienceDirect (2025).

IoT Analytics Using Published IoT Datasets

1. Kumar, V. et al., “Sense (and) the city: From Internet of Things sensors and open data platforms tourban observatories,” *IET Smart Cities*, Wiley, vol. 6, no. 4, pp. 291–311 (2024).
2. Rabbani, M. et al., “Device Identification and Anomaly Detection in IoT Environments,” *IEEE Internet ofThings Journal* (Dec. 2024) — introduces and uses the CIC IoT-DIAD 2024 dataset.
3. Neto, E. C., Dadkhah, S., Ferreira, R., Zohourian, A., Lu, R. and Ghorbani, A. A., “CICIoT2023: Areal-time dataset and benchmark for large-scale attacks in IoT environment,” *Sensors*, MDPI (2023).
4. Adjei, J., Zincir-Heywood, N., Nandy, B. and Seddigh, N., “Dalhousie NIMS Lab IoT 2024 Dataset,” *IEEE DataPort* (2024), doi: 10.21227/shzy-z570.

IoT Security Framework

1. Khan, M. A. et al., “A Deep Learning-Based Intrusion Detection System for MQTT Enabled IoT,” *Sensors*, MDPI, vol. 21, no. 21, Art. 7016 (2023), doi: 10.3390/s21217016.
2. “Enhancing MQTT security for internet of things: Lightweight two-way authorization and authentication with advanced security measures,” *Measurement: Sensors*, ScienceDirect (2024).
3. “An efficient intrusion detection system for MQTT-IoT using enhanced chaotic salp swarm algorithm and LightGBM,” *International Journal of Information Security*, Springer (2023).

Predictive Maintenance Using Industrial IoT Data

1. Aboshosha, A., Haggag, A., George, N. and Hamad, H. A., “IoT-based data-driven predictive maintenance relying on fuzzy system and artificial neural networks,” *Scientific Reports*, Nature, doi: 10.1038/s41598-023-38887-z (2023).
2. “Detecting Adversarial Attacks in IoT-Enabled Predictive Maintenance with Time-Series Data Augmentation,” *Information*, MDPI, vol. 15, no. 11, p. 740 (2024).
3. Özcan, H., “Predictive maintenance in aircraft engine maintenance using the C-MAPSS dataset: performance comparison and evaluation of machine learning classification algorithms,” Cambridge University Press (2026).
4. “IoT-driven Predictive Maintenance for Enhanced Reliability in Industrial Applications,” IEEE Conference Publication, IEEE Xplore (2025).

1. Edge Computing Framework for IoT

Objective: Design and implement an edge computing framework using Raspberry Pi to process IoT data locally and compare its performance with cloud-based processing.

2. MQTT Performance Analysis

Objective: Analyze the performance of the MQTT protocol under different Quality of Service (QoS) levels and varying workloads.

3. Raspberry Pi IoT Gateway

Objective: Develop an IoT gateway on Raspberry Pi that supports protocol translation between MQTT, HTTP, CoAP, and WebSocket.

4. IoT Intrusion Detection System

Objective: Design an intrusion detection system for IoT networks to identify cyber attacks such as DoS, replay attacks, unauthorized access, and port scanning.

5. Docker-Based Edge Computing Platform

Objective: Build a containerized edge computing platform on Raspberry Pi using Docker for deploying IoT services such as MQTT Broker, Node-RED, Grafana, and InfluxDB.

6. Federated Learning on Raspberry Pi

Objective: Implement a federated learning framework where multiple Raspberry Pi devices collaboratively train machine learning models while preserving data privacy.

7. TinyML Benchmarking on Raspberry Pi

Objective: Evaluate the performance of lightweight AI inference frameworks such as TensorFlow Lite, ONNX Runtime, and PyTorch for edge AI applications.

8. Smart Home Digital Twin

Objective: Develop a digital twin of a smart home using Raspberry Pi to simulate and monitor virtual IoT devices in real time.

9. Raspberry Pi Cluster for Edge Computing

Objective: Build a Raspberry Pi cluster to study distributed computing, load balancing, fault tolerance, and task scheduling for edge applications.

10. Edge AI Server

Objective: Develop an edge AI server on Raspberry Pi capable of executing AI inference models for real-time intelligent applications.

11. Secure MQTT Broker

Objective: Implement a secure MQTT communication platform with authentication, TLS encryption, access control, and performance evaluation.

12. Software-Defined IoT Network

Objective: Design and evaluate a Software-Defined Networking (SDN)-based IoT architecture for efficient traffic management and Quality of Service (QoS).

13. Blockchain-Based IoT Security

Objective: Develop a blockchain-enabled IoT platform to provide secure, decentralized, and tamper-resistant communication among IoT devices.

14. AI-Based Predictive Maintenance

Objective: Develop an AI-driven predictive maintenance system using publicly available industrial datasets to detect faults and predict equipment failures without physical sensors.